

SPA con React

5. Librerías de Componentes



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Librerías de componentes

- React **no proporciona** componentes de alto nivel, permite usar HTML, CSS para dar estilo a las páginas
- Existen **diferentes librerías** que facilitan el diseño de páginas web:
 - Frameworks CSS
 - Librerías genéricas JavaScript
 - Librerías que se integran en React
 - Librerías con múltiples componentes
 - Librerías de charts (Gráficas)
 - Librerías de subida de ficheros
 - ...

Librerías de componentes

- Frameworks CSS
 - Se puede usar cualquier framework de componentes que **sólo tenga CSS**



- **Usar Bootstrap CSS**

- Instalamos la dependencia de Bootstrap 5 con NPM

```
npm install --save bootstrap
```

- Se importa el CSS en el home.tsx (o root.tsx)

```
import 'bootstrap/dist/css/bootstrap.min.css';

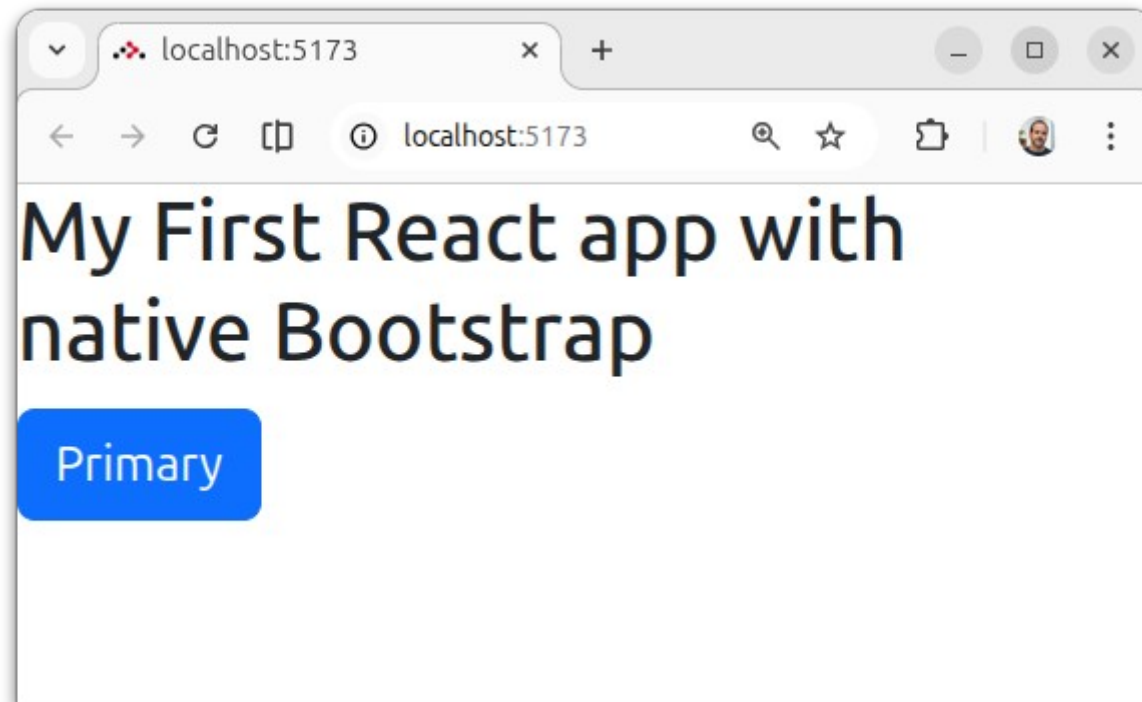
export default function Home() {

  return (
    <>
      <h1>My First React app with native Bootstrap</h1>
      <button type="button" className="btn btn-primary">Primary</button>
    </>
  );
}
```

Librerías de componentes

exam34_bootstrap_css

- Usar Bootstrap CSS



- Librerías de componentes JavaScript
 - El uso de librerías gráficas nativas JavaScript puede resultar complejo (por la integración con el sistema de renderizado de React)
 - Aunque la librería **funcione**, puede ser **ineficiente** porque puede “colisionar” con la librería a la hora de modificar el DOM
 - Es mejor usar **adaptadores de las librerías** para React

Librerías de componentes

- **React bootstrap**
 - Reimplementación de la parte JavaScript de Bootstrap basada en React



<https://react-bootstrap.netlify.app/>

- Están implementados la mayoría de los componentes Bootstrap
 - Alert
 - Buttons
 - Carousel
 - Collapse
 - Datepicker
 - Dropdown
 - Modal
 - Pagination
 - Popover
 - Progressbar
 - Rating
 - Tabs
 - Accordion
 - Timepicker
 - Tooltip

React bootstrap

- Instalación

- Instalar las dependencias NPM

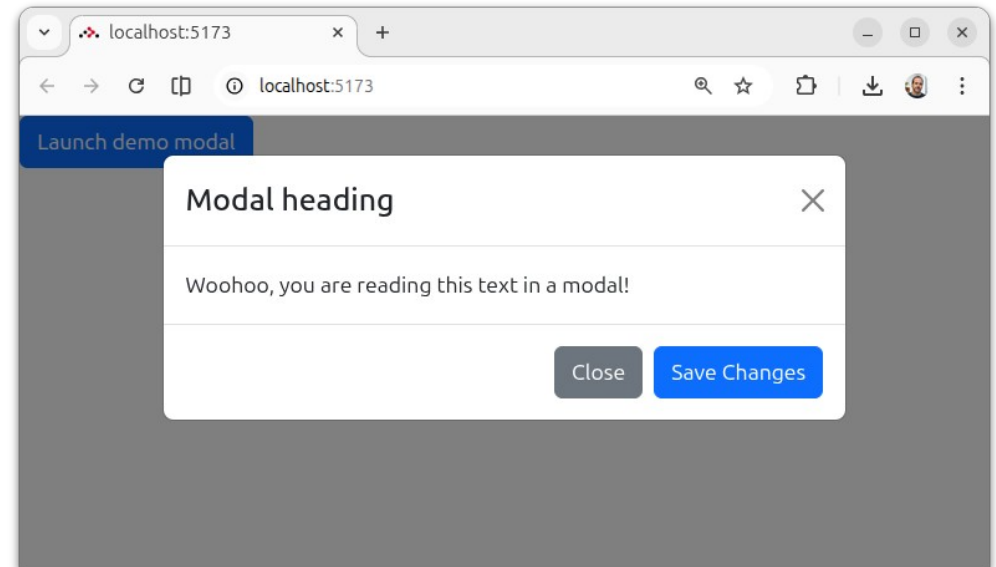
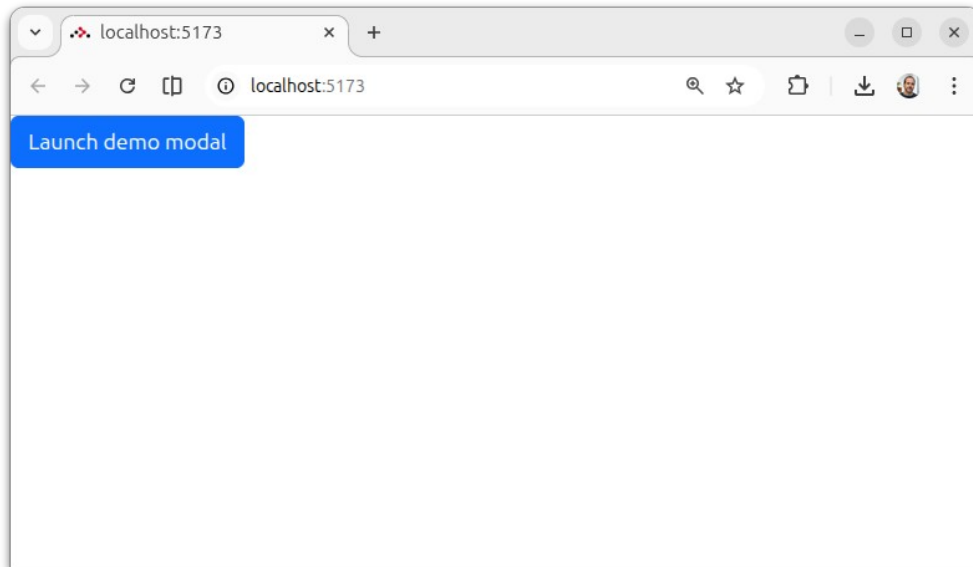
```
npm install react-bootstrap bootstrap
```

- Uso de las componentes React

```
import 'bootstrap/dist/css/bootstrap.min.css';
import Button from 'react-bootstrap/Button';
import Modal from 'react-bootstrap/Modal';
...
export default function Home() {
  ...
  return (
    <Button variant="primary" ...>
      Push me!
    </Button>
  );
}
```

React bootstrap

exam35_bootstrap_modal



React bootstrap

exam35_bootstrap_modal

```
import 'bootstrap/dist/css/bootstrap.min.css';
import { useState } from 'react';
import Button from 'react-bootstrap/Button';
import Modal from 'react-bootstrap/Modal';

export default function Home() {

  const [show, setShow] = useState(false);

  const handleClose = () => setShow(false);
  const handleShow = () => setShow(true);

  return (
    <>
      <Button variant="primary" onClick={handleShow}>
        Launch demo modal
      </Button>

      <Modal show={show} onHide={handleClose}>
        <Modal.Header closeButton>
          <Modal.Title>Modal heading</Modal.Title>
        </Modal.Header>
        <Modal.Body>
          Woohoo, you are reading this text in a modal!
        </Modal.Body>
        <Modal.Footer>
          <Button variant="secondary" onClick={handleClose}>
            Close
          </Button>
          <Button variant="primary" onClick={handleClose}>
            Save Changes
          </Button>
        </Modal.Footer>
      </Modal>
    </>
  );
}
```

React bootstrap

exam35_bootstrap_modal

```
import 'bootstrap/dist/css/bootstrap.min.css';
import { useState } from 'react';
import Button from 'react-bootstrap/Button';
import Modal from 'react-bootstrap/Modal';

export default function Home() {

  const [show, setShow] = useState(false);

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          <Button variant="secondary" onClick={handleClose}>
            Close
          </Button>
          <Button variant="primary" onClick={handleClose}>
            Save Changes
          </Button>
        </Modal.Footer>
      </Modal>
    </>
  );
}
```

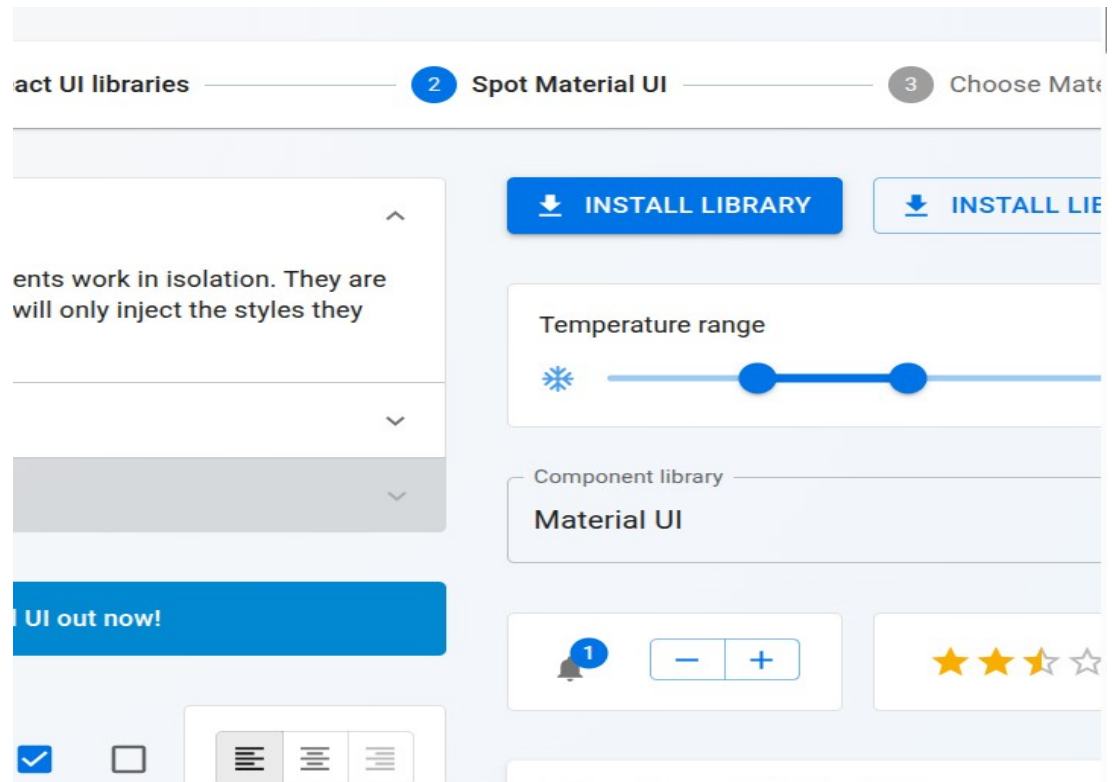
Se usan los
componentes React

Librerías de componentes

- **Material UI**
 - Librería de componentes con el estilo Material Design de Google para React



<https://mui.com/material-ui/>



- ## Instalación

- Instalar dependencias NPM

```
npm install @mui/material @emotion/react @emotion/styled
```

```
npm install @fontsource/roboto
```

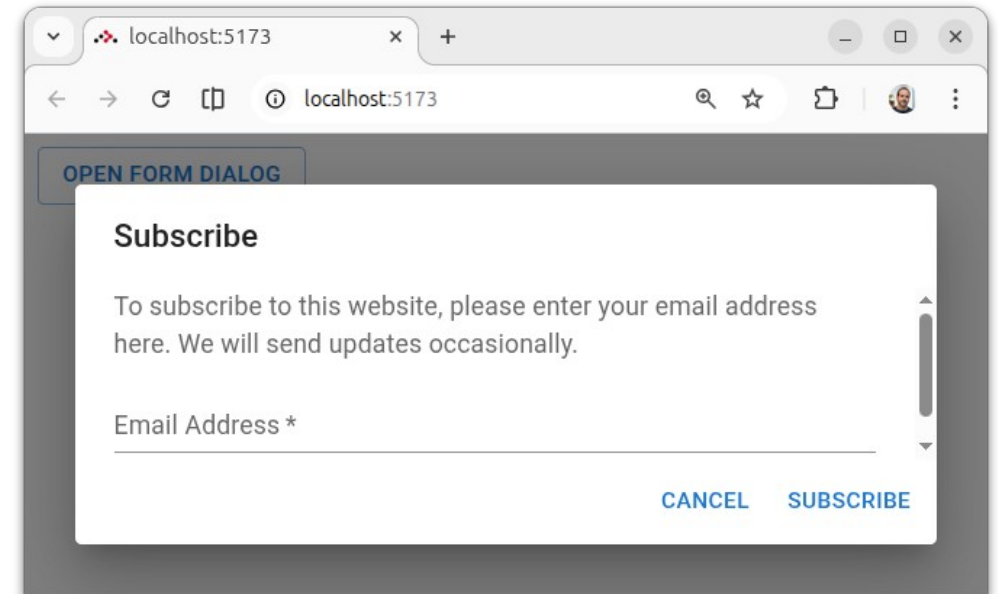
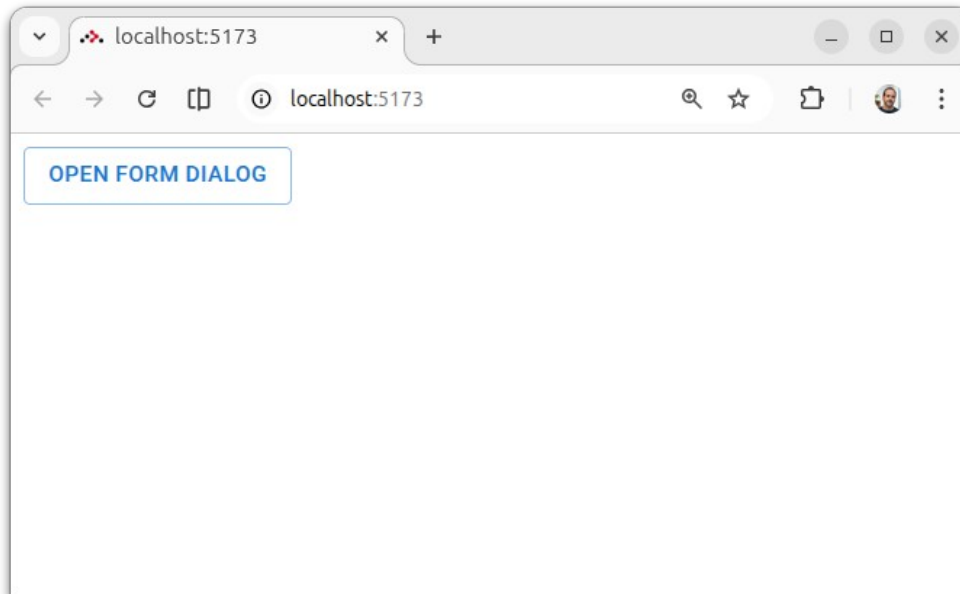
```
npm install @mui/icons-material
```

- CSS

```
import '@fontsource/roboto/300.css';  
import '@fontsource/roboto/400.css';  
import '@fontsource/roboto/500.css';  
import '@fontsource/roboto/700.css';
```

Material UI

exam36_material



Material UI

exam36_material

```
import '@fontsource/roboto/300.css';
import '@fontsource/roboto/400.css';
import '@fontsource/roboto/500.css';
import '@fontsource/roboto/700.css';

import Button from '@mui/material/Button';
import TextField from '@mui/material/TextField';
import Dialog from '@mui/material/Dialog';
import DialogActions from '@mui/material/DialogActions';
import DialogContent from '@mui/material/DialogContent';
import DialogContentText from '@mui/material/DialogContentText';
import DialogTitle from '@mui/material/DialogTitle';
import { useState, type SubmitEvent } from 'react';

export default function Home() {

  const [open, setOpen] = useState(false);

  const handleClickOpen = () => { setOpen(true); };
  const handleClose = () => { setOpen(false); };

  const handleSubmit = (event: SubmitEvent) => {
    event.preventDefault();
    const formData = new FormData(event.target);
    const formJson = Object.fromEntries((formData as any).entries());
    const email = formJson.email;
    console.log(email);
    handleClose();
  };

  return ( ... )
}
```

Material UI

exam36_material

```
<>
<Button variant="outlined" onClick={handleClickOpen}>
  Open form dialog
</Button>
<Dialog open={open} onClose={handleClose}>
  <DialogTitle>Subscribe</DialogTitle>
  <DialogContent>
    <DialogContentText>
      To subscribe to this website, please enter your email address here. We
      will send updates occasionally.
    </DialogContentText>
    <form onSubmit={handleSubmit} id="subscription-form">
      <TextField
        autoFocus
        required
        margin="dense"
        id="name"
        name="email"
        label="Email Address"
        type="email"
        fullWidth
        variant="standard"
      />
    </form>
  </DialogContent>
  <DialogActions>
    <Button onClick={handleClose}>Cancel</Button>
    <Button type="submit" form="subscription-form">Subscribe</Button>
  </DialogActions>
</Dialog>
</>
```

Librerías de componentes

- Existen librerías que proporcionan un componente concreto (o de la misma familia)
- ag-grid: Tabla con controles avanzados



<https://www.ag-grid.com/>

Librerías de componentes

Demo - Performance Grid

ag-grid.com/example/

AG Grid

Products

Search

Ctrl K

Demos

Theme Builder

Docs

API

Community

Pricing

Performance

See on GitHub

See the video tour

Example showing grid performance with adjustable rows and columns.

Data Size: 100 Rows, 22 Cols Theme: Quartz Filter: Filter any column...

Drag here to set row groups

	Participant	Game of Choice	Performance	Rating	Total Winnings	Monthly Breakdown			
	Name	Language	Country	Game Name	Bought	Bank Balance	Jan		
1	Tony Smith	English	Ireland	Chess	✓	\$2397	★★	\$569.571	\$38.03
2	Andrew Connell	Swedish	Sweden	Bul	✓	\$12.749	★★★	\$481.734	\$17.69
3	Kevin Flanagan	Spanish	Uruguay	Rithmomachy	□	\$95.078		\$747.956	\$54.89
4	Bricker McGee	French	France	Kalah	□	\$65.506		\$605.384	\$81.08
5	Dimple Unalkat	Portuguese	Portugal	Game of the Generals	✓	\$85.310	★★	\$600.036	\$73.94
6	Gil Lopes	Spanish	Colombia	Hare and Hounds	✓	\$75.701	★★	\$574.681	\$20.47
7	Sophie Beckham	English	Ireland	Sugoroku	□	\$66.706		\$651.234	\$44.44
8	Isabelle Black	French	France	Nine Men's Morris	□	\$15.749	★★★	\$497.221	\$32.41
9	Emily Braxton	Maltese	Malta	Blockade	✓	\$4057	★★★★★	\$622.755	\$82.94
10	Olivia Brennan	French	France	Patolli	✓	\$32.835	★	\$727.405	\$93.51
11	Lily Brock	Italian	Italy	YINSH	□	\$7440	★★★★★	\$563.168	\$34.46

Rows: 100

Pivot Mode

Search...

Participant

Name

Language

Country

Game of Choice

Game Name

Bought

Performance

Bank Balance

Rating

Total Winnings

Monthly Breakdown

Row Groups

Drag here to set row groups

Values

avg(Bank Balance)

avg(Rating)

sum(Total Winnings)

Librerías de componentes

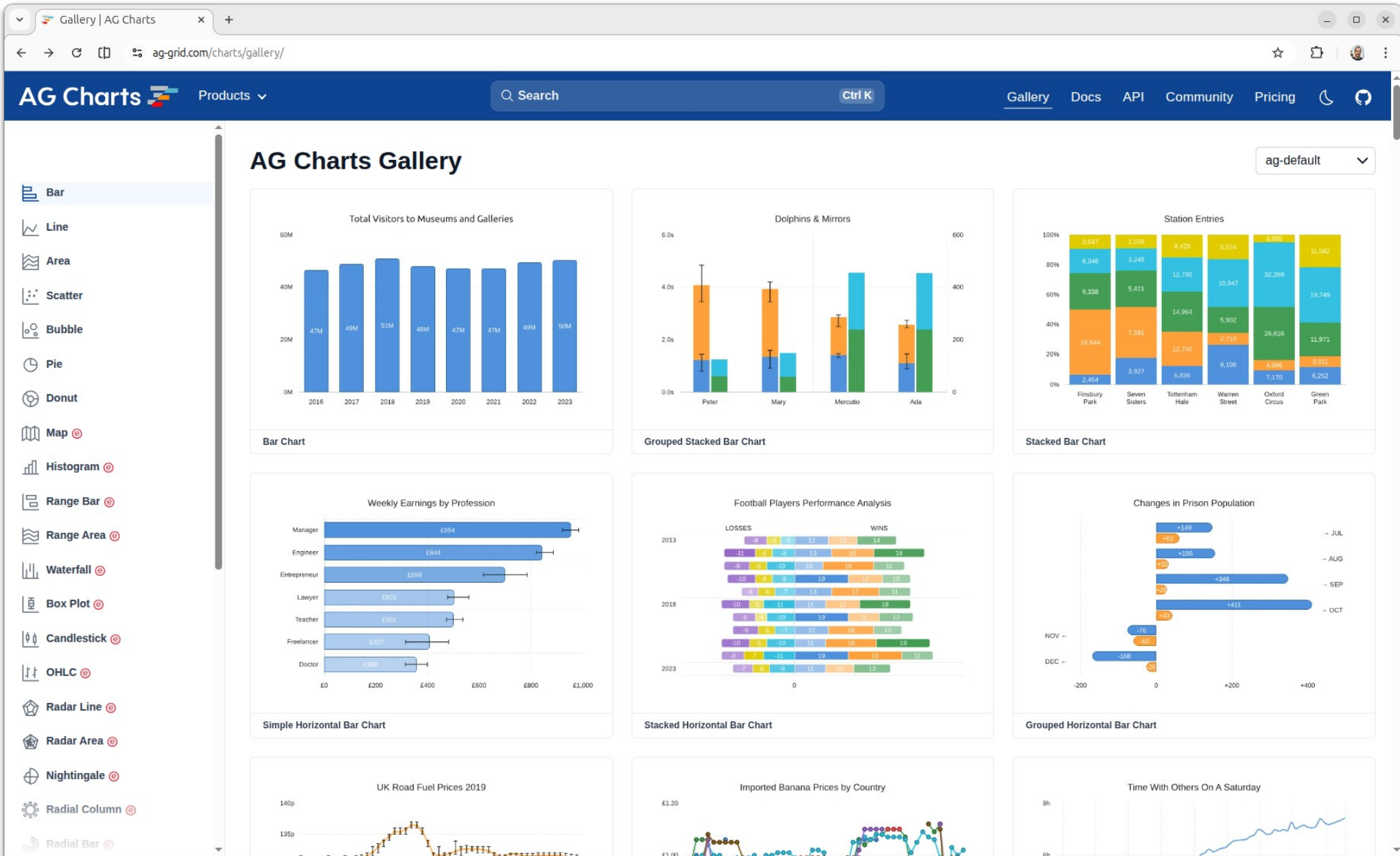
- ag-charts: Gráficos de barras, líneas, etc...

AG Charts



<https://www.ag-grid.com/charts/>

Librerías de componentes



Librerías de componentes

- Plantillas de aplicación
 - Como las **plantillas web**, pero para React
 - Lo más habitual son los “paneles de **administración**”
 - Los de mayor calidad suelen tener un **coste económico**
 - Al seleccionar una plantilla hay que tener en cuenta que esté diseñada para React 19 y React Router
 - Si se han creado para otro framework se tienen que adaptar

Librerías de componentes

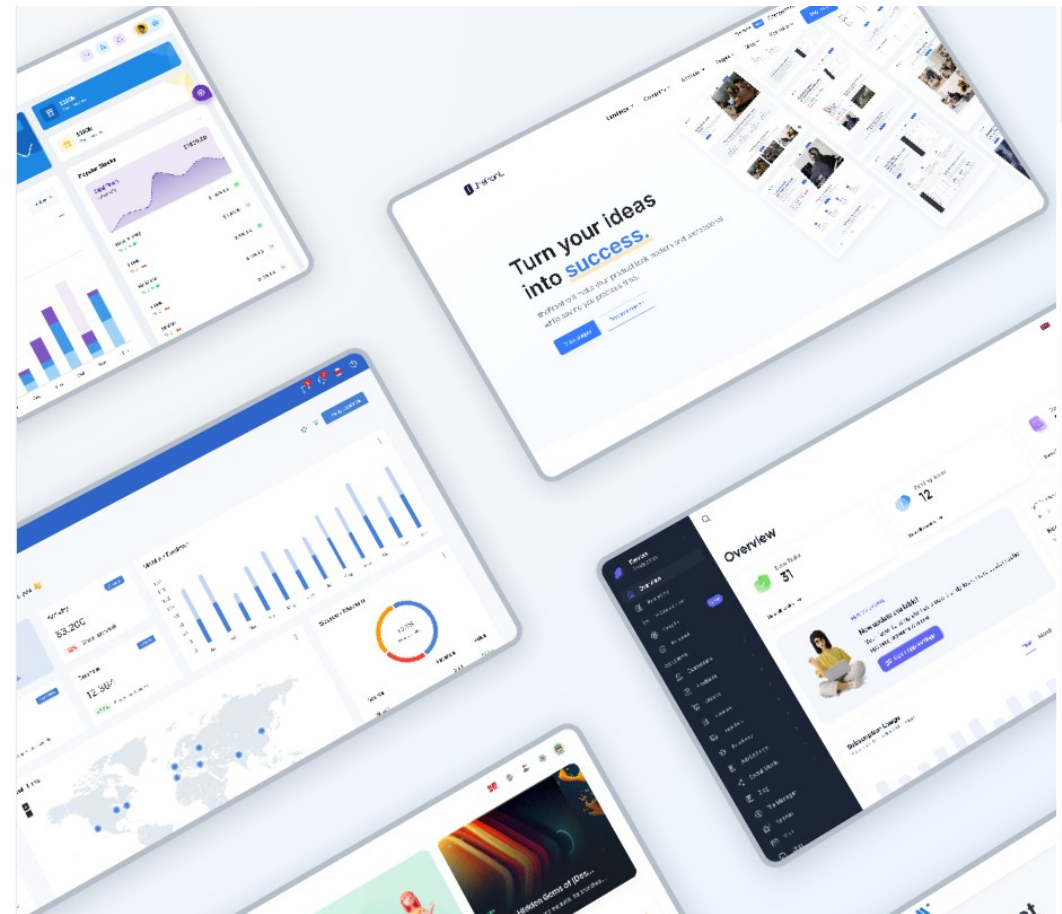
- MUI Templates

Templates

Beautiful and fully built Material UI templates

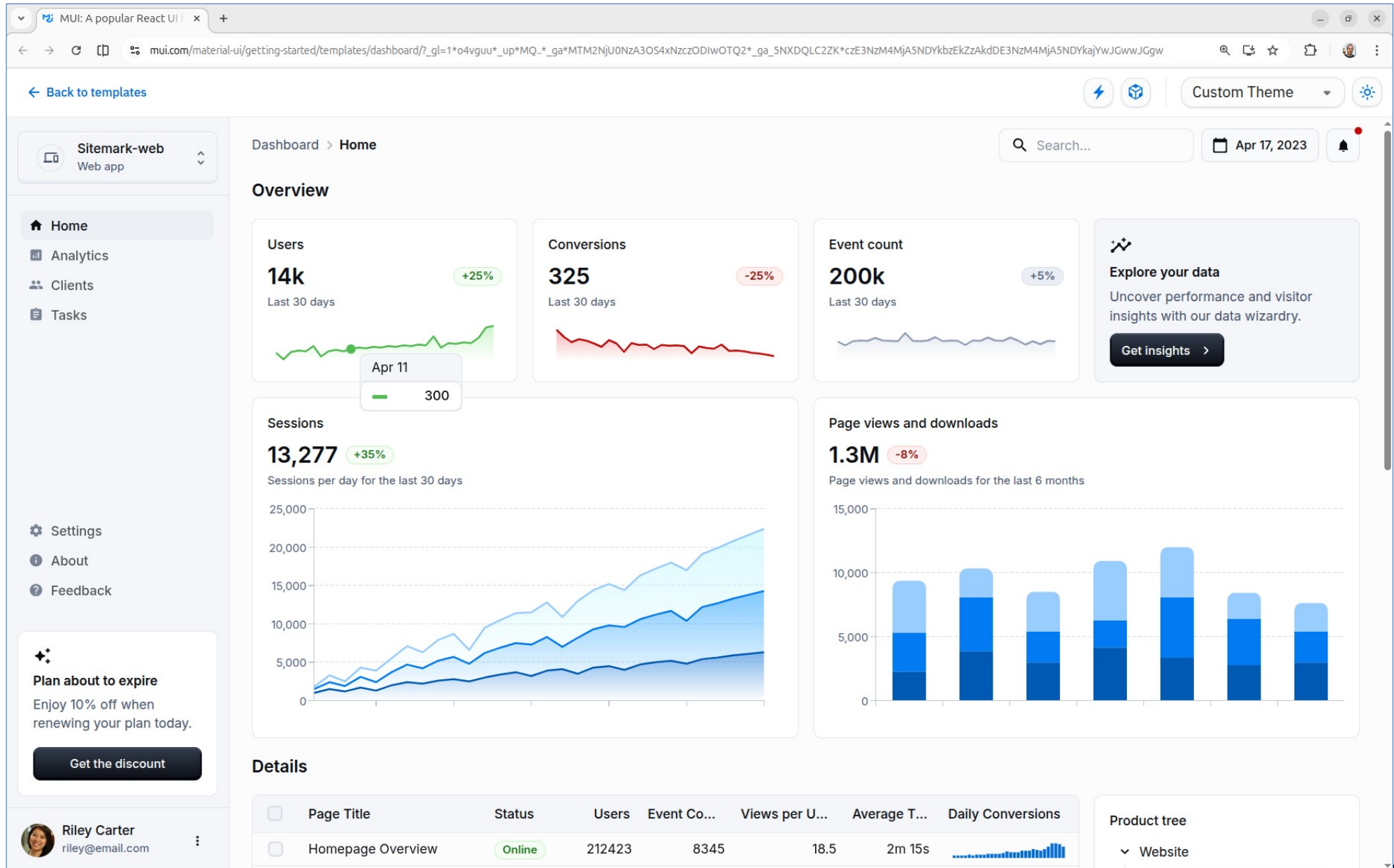
A collection of 4.5 average rating templates, selected and curated by Material UI's maintainers to get your projects up and running today.

[Browse templates >](#)



<https://mui.com/templates/>

MUI Templates



Librerías de componentes

- Prime React



<https://primereact.org/>

Prime React

The screenshot displays the PrimeReact Sakai dashboard. The browser address bar shows 'sakai.primereact.org'. The dashboard features a dark theme with a sidebar on the left containing a 'HOME' section with a 'Dashboard' link and a 'UI COMPONENTS' section listing various UI elements like Form Layout, Input, Float Label, Invalid State, Button, Table, List, Tree, Panel, Overlay, Media, Menu, Message, File, Chart, and Misc. Below these are 'PRIME BLOCKS' (Free Blocks, All Blocks) and 'UTILITIES' (Primalone). The main content area includes four summary cards: Orders (152, 24 new since last visit), Revenue (\$2,100, 52% increase since last week), Customers (28,441, 520 newly registered), and Comments (152 Unread, 85 responded). A 'Recent Sales' table lists items like Bamboo Watch, Black Watch, Blue Band, Blue T-Shirt, and Bracelet with their prices and view counts. A 'Sales Overview' chart shows two datasets (First and Second) over a seven-month period. A 'Notifications' section lists recent transactions and user actions.

HOME

- Dashboard

UI COMPONENTS

- Form Layout
- Input
- Float Label
- Invalid State
- Button
- Table
- List
- Tree
- Panel
- Overlay
- Media
- Menu
- Message
- File
- Chart
- Misc

PRIME BLOCKS

- Free Blocks
- All Blocks

UTILITIES

- Primalone

Summary Cards:

- Orders:** 152 (24 new since last visit)
- Revenue:** \$2,100 (%52+ since last week)
- Customers:** 28441 (520 newly registered)
- Comments:** 152 Unread (85 responded)

Recent Sales Table:

Image	Name ↑↓	Price ↑↓	View
	Bamboo Watch	\$65.00	View
	Black Watch	\$72.00	View
	Blue Band	\$79.00	View
	Blue T-Shirt	\$29.00	View
	Bracelet	\$15.00	View

Sales Overview Chart:

Line chart showing Sales Overview for First Dataset (blue) and Second Dataset (green) from January to July. The Y-axis ranges from 10 to 90.

Notifications:

TODAY

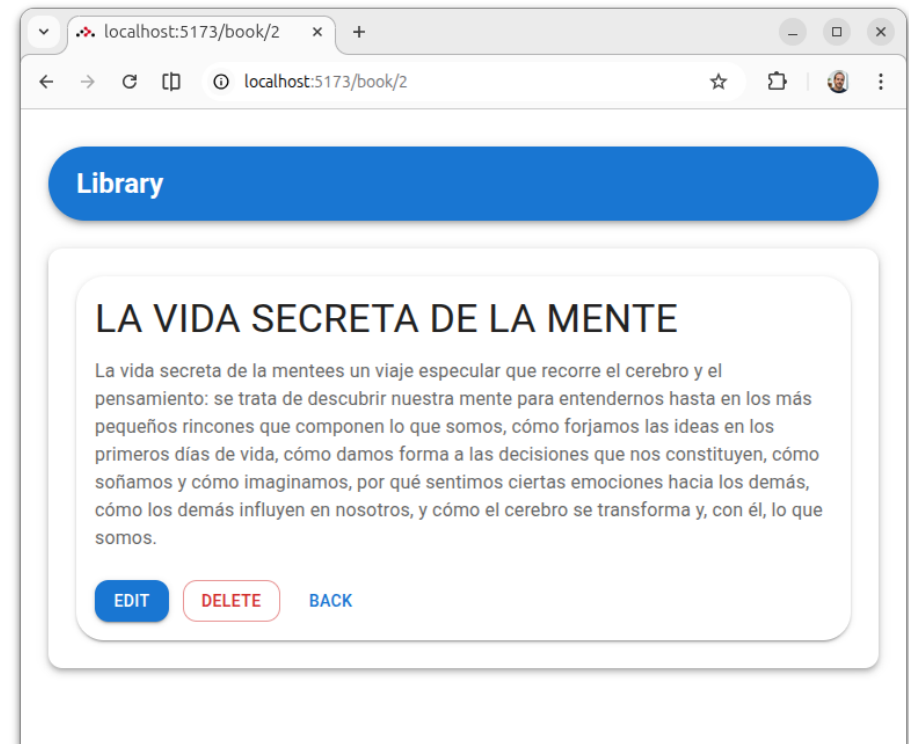
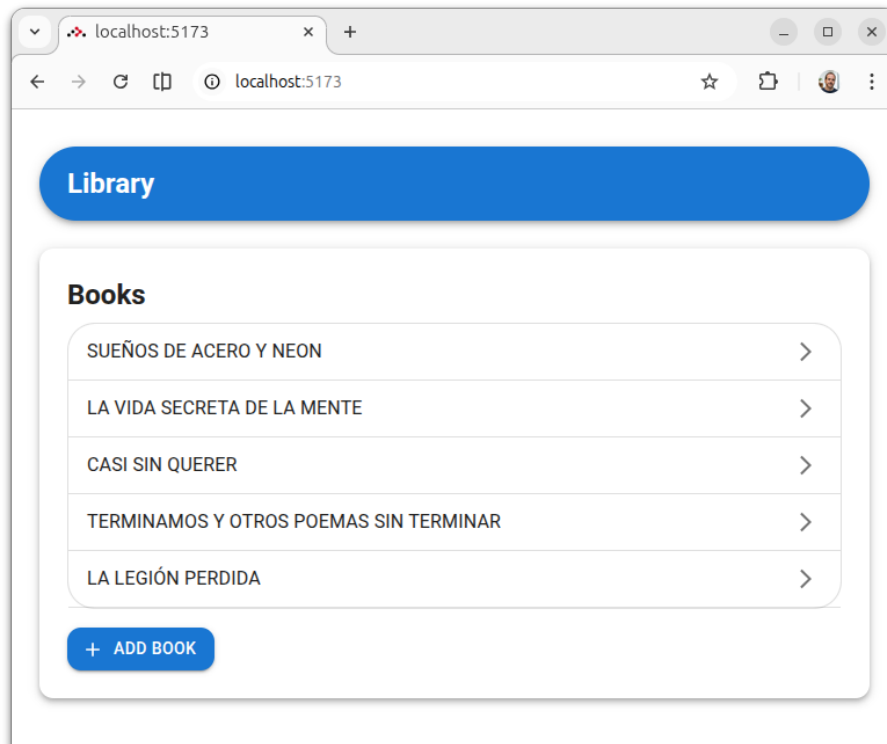
- Richard Jones has purchased a blue t-shirt for 79\$
- Your request for withdrawal of 2500\$ has been initiated.

YESTERDAY

- Keyser Wick has purchased a black jacket for 59\$
- Jane Davis has posted a new questions about your product.

Ejercicio 11

- Mejora el aspecto de la aplicación de gestión de libros usando una librería de componentes
- Añade un diálogo de confirmación al borrar un libro



Librerías de componentes

- Existen **librerías de componentes** que mejoran el estilo y el comportamiento de los componentes nativos del navegador
- Existen **librerías de componentes** especializados (tablas, gráficos, etc.)
- Existen **plantillas (*templates*)** de aplicaciones completas que simplifican la creación de aplicaciones